

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
1	(a)	(i)		Any two × (1) from: Flow <u>rate</u> /speed/current (1) temperature (1) (named) {ion/mineral} <u>concentration</u> (1) NOT content conductivity (1) oxygen <u>concentration</u> (1) light <u>intensity/exposure</u> (1) vegetation {on banks/in water}(1) Accept algae in water time of year/season/time of day (1) Reject pH/width/depth/stream bed/any ref to method/technique		2		2		2
		(ii)		Any two × (1) from: A. (same) sampling area (1) NOT distance B. {time spent/force/power/speed/amount} <u>of kicking</u> /number of kicks (1) C. net mesh size (1) NOT area of net/same net D. Distance from bank/distance of kicking from the net (1) Reject any reference to stream conditions/time net is in water		2		2		2
		(iii)		would <u>kill</u> them/animals die(1) preserve diversity/effect on food chains/endangered species(1) Accept reverse answers for riverside identification		2		2		2
	(b)	(i)		Remove {bias/experimenter choice} (1) Reject fair test/accuracy/reliability/validity	1			1		1

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		(ii)		Any two × (1) from: A. Animals not caught by net/ Missed net/passed through net/net removed too soon (1) B. Animals not dislodged from stream bed/not kicked hard enough to dislodge them/organisms moved away from sample area (1) C. Not kicked entire area/uneven kicking (1) D. Unable to identify animals/animals too small to identify (1)		2		2		2
		(iii)		$D = 0.7/0.699 = 3$ marks <i>If incorrect rounding e.g. 0.69 award 2 marks</i> <i>If incorrect answer award</i> $D = 0.3/0.301 = 2$ marks OR $N(N-1) = 6972$ (1) $\Sigma n(n-1) = 2098$ (1)		3		3	3	
		(iv)		Forest stream: Lower <u>species</u> richness/fewer <u>species</u> (1) Accept correct use of numbers Lower <u>species</u> evenness/dominated by {single (named) <u>species</u> /stonefly nymph} (1) ORA moorland stream			2	2		
	(c)	(i)		More (than five) sample sites/{more/larger} area sampled(1) More (than two) streams (1) Not forest unqualified			2	2		2
		(ii)		Create buffer zone between forest and stream/cut down trees next to stream/{ <u>increase pH of</u> /neutralise} stream (eg liming)/OWTTE (1) NOT cut down trees Introduction of more species = neutral			1	1		1
				Question 1 total	1	11	5	17	3	12